

SAT Unit 1.1

Linear Equations in One Variable

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- 2 Three Types of Solutions
- 3 Real World Application
- 4 Practice

Solving for Linear Equation

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Introduction to Linear Equations

Definition: A **linear equation in one variable** is an equation that can be written in the form:

$$ax + b = 0$$

where a and b are constants, and x is the variable.

Key Characteristics:

- The highest exponent of x is 1, making it a **first-degree** equation.
- The solution to a linear equation is the value of x that makes the equation true.
- The equation represents a straight line when graphed on the coordinate plane.

Example: Solve for x

Solve the equation:

$$2x - 5 = 9$$

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Solution:

$$2x - 5 = 9$$

$$2x = 14$$

$$x = 7$$

Three Types of Solutions

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- **One Solution:** The equation simplifies to $x = c$, where c is a constant.

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- **Infinite Solutions:** The equation simplifies to an identity, such as $0 = 0$, meaning any x is a solution. Example:

$$4x + 2 = 2(2x + 1) \Rightarrow 4x + 2 = 4x + 2$$

Example 3: No Solution Case

Consider the equation:

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Given that the equation has no solution, find k .

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Solution:

$$3kx + 39 = \frac{48}{17}x + 36$$

$$x(3k - \frac{48}{17}) = -3$$

Since there is no solution, the coefficient of x must be zero:

$$3k - \frac{48}{17} = 0$$

Solving for k :

$$k = \frac{16}{17}$$

Algebra Problem: No Solution Condition

Problem Statement:

Given the equation:

$$2(kx - n) = \left(-\frac{28}{15}\right)x - \frac{36}{19}$$

where k and n are constants, and $n > 1$. The equation has no solution.

Question: What is the value of k ?

Solution: Finding k

Step 1: Expand both sides

Expanding the left-hand side:

$$2kx - 2n$$

The right-hand side remains:

$$-\frac{28}{15}x - \frac{36}{19}$$

Step 2: Compare coefficients of x

Since the equation has ****no solution****, the coefficients of x must be equal, but the constants must create a contradiction.

$$2k = -\frac{28}{15}$$

Step 3: Solve for k

$$k = \frac{-28}{30} = \frac{-14}{15}$$

Question

Consider the equation:

$$9x + 5 = a(x + b)$$

where a and b are constants. The equation has no solutions.

Which of the following must be true?

- ① $a = 9$
- ② $b = 5$
- ③ $b \neq \frac{5}{9}$

Answer Choices:

- A. None
- B. I only
- C. I and II only
- D. I and III only

Solution

Step 1: Understanding the Equation

- The given equation is:

$$9x + 5 = a(x + b)$$

- For the equation to have **no solution**, both sides must be parallel, meaning:

$$9 = a, \quad b \neq \frac{5}{9}$$

Step 2: Answer Choices

- $a = 9$ must be true (Statement I is correct).
- $b = 5$ is not necessarily true (Statement II is incorrect).
- $b \neq \frac{5}{9}$ must be true (Statement III is correct).

Final Answer: Option **D (I and III only)**.

Real World Application

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Word Problem: Corn Removal Rate

Problem Statement:

Hector used an auger to remove corn from a storage bin at a constant rate. The bin contained 24,000 bushels of corn when Hector began to use the auger. After 5 hours, 19,350 bushels remained in the bin. If the auger continues to remove corn at this rate, what is the total number of hours Hector will have been using the auger when 12,840 bushels remain?

Answer Choices:

- A 3
- B 7
- C 8
- D 12

Solution: Finding the Total Hours

Step 1: Find the rate of corn removal

- Initial amount of corn: 24,000 bushels - Remaining after 5 hours: 19,350 bushels - Corn removed in 5 hours:

$$24,000 - 19,350 = 4,650 \text{ bushels}$$

- Rate of removal:

$$\frac{4,650}{5} = 930 \text{ bushels per hour}$$

Step 2: Find the total hours needed for 12,840 bushels remaining

- Find the total corn removed:

$$24,000 - 12,840 = 11,160 \text{ bushels}$$

- Find the total hours:

$$\frac{11,160}{930} = 12 \text{ hours}$$

Final Answer: D. 12

Glenview Street Property Discount

Table: Townsend Realty Group Investments

Property address	Purchase price (\$)	Monthly rental price (\$)
Clearwater Lane	128,000	950
Driftwood Drive	176,000	1,310
Edgemont Street	70,000	515
Glenview Street	140,000	1,040
Hamilton Circle	450,000	3,365

Townsend Realty Group purchased the Glenview Street property and received a 40% discount off the original price along with an additional 20% off the *discounted* price for purchasing in cash.

Which of the following best approximates the **original price**, in dollars, of the Glenview Street property?

- A. \$350,000
- B. \$291,700
- C. \$233,300
- D. \$175,000

Solution: Glenview Street Property

Let P be the original price of the Glenview Street property.

- First, a 40% discount was applied:

$$P \times 0.60$$

- Then, an additional 20% discount was applied to the new value:

$$P \times 0.60 \times 0.80 = 0.48P$$

- We are told the final purchase price was \$140,000:

$$0.48P = 140,000 \Rightarrow P = \frac{140,000}{0.48} = 291,666.67$$

Answer: B. 291,700

Practice

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Question : Infinitely Many Solutions

Given:

$$\frac{12x + 28}{4} - \frac{s}{13} = r(x - 8)$$

where s and r are constants, and $s > 0$.

If the equation has infinitely many solutions, what is the value of s ?

Answer: Infinitely Many Solutions

Step 1: Simplify the left side

$$\frac{12x + 28}{4} = 3x + 7 \Rightarrow 3x + 7 - \frac{s}{13}$$

So the equation becomes:

$$3x + 7 - \frac{s}{13} = r(x - 8)$$

Step 2: Expand the right side

$$r(x - 8) = rx - 8r$$

Step 3: For infinitely many solutions, both sides must be identical

$$\text{Match coefficients: } 3 = r \Rightarrow r = 3$$

$$\begin{aligned}\text{Match constants: } 7 - \frac{s}{13} &= -8r = -24 \\ \Rightarrow \frac{s}{13} &= 31 \Rightarrow s = 403\end{aligned}$$

Correct Answer: $s = 403$

Question: No Solution Linear Equation

Given the equation:

$$\frac{1}{3}(bx + c) = \frac{14}{27}x + \frac{4}{9}$$

where b and c are constants and $c > 3$.

The equation has **no solution**. What is the value of b ?

Answer: No Solution Condition

Step 1: Expand the left-hand side:

$$\frac{1}{3}(bx + c) = \frac{b}{3}x + \frac{c}{3}$$

Now the equation becomes:

$$\frac{b}{3}x + \frac{c}{3} = \frac{14}{27}x + \frac{4}{9}$$

Step 2: For a linear equation to have no solution, the coefficients of x must be equal but the constants must be different.

Match the coefficients:

$$\frac{b}{3} = \frac{14}{27} \Rightarrow b = \frac{14}{27} \cdot 3 = \frac{14}{9}$$

Check the constants:

$$\frac{c}{3} \neq \frac{4}{9}, \quad \text{and } c > 3 \Rightarrow \text{This guarantees different constants}$$

Question: No Solution for Linear Equation

Given the equation:

$$ax - b = 3(2x + 1)$$

where a and b are constants, and the equation has **no solution**. Which of the following could be the values of a and b ?

A. $a = 2, b = -3$

B. $a = 2, b = 3$

C. $a = 6, b = -3$

D. $a = 6, b = 3$

Answer: No Solution Condition

Step 1: Expand the right-hand side

$$ax - b = 6x + 3$$

For no solution: Coefficients of x must match, but constants must not:

$$a = 6, \quad -b \neq 3 \Rightarrow b \neq -3$$

Now test options:

A. $a = 2$, not valid

B. $a = 2$, not valid

C. $a = 6, b = -3 \rightarrow -b = 3$, so both sides match $\rightarrow$ *infinite* solutions

D. $a = 6, b = 3 \rightarrow -b = -3 \neq 3$, **valid: no solution**

Correct Answer: D

Question: Absolute Value Equation

Solve for the **positive solution** to the equation:

$$3|5 - x| = 42 - 4|5 - x|$$

Answer: Solving Absolute Value Equation

Step 1: Let $y = |5 - x|$, then:

$$3y = 42 - 4y \Rightarrow 7y = 42 \Rightarrow y = 6$$

Step 2: Solve $|5 - x| = 6$

$$5 - x = 6 \Rightarrow x = -1 \quad \text{or} \quad 5 - x = -6 \Rightarrow x = 11$$

Step 3: Positive solution: 11

Question: Popcorn Word Problem

Joey and Matt were tasked with bringing popcorn to a certain event. Joey brought x pounds of popcorn and Matt brought 4 fewer pounds than Joey did.

During the event, 60% of the popcorn they had brought was consumed. If Joey and Matt divided the leftover popcorn evenly between them, which of the following represents the number of pounds of leftover popcorn each of them took?

- A. $0.2x - 0.8$
- B. $0.4x - 0.8$
- C. $0.6x - 1.2$
- D. $0.8x - 1.6$

Answer: Popcorn Division

Step 1: Total popcorn brought

$$\text{Joey: } x, \quad \text{Matt: } x - 4 \Rightarrow \text{Total: } x + (x - 4) = 2x - 4$$

Step 2: 60% consumed $\rightarrow$ 40% remaining

$$\text{Leftover: } 0.4(2x - 4) = 0.8x - 1.6$$

Step 3: Divided evenly between 2 people

$$\frac{0.8x - 1.6}{2} = 0.4x - 0.8$$

Final Answer: B. $0.4x - 0.8$

Summary

- A linear equation in one variable has **one solution**, **no solution**, or **infinitely many solutions**.
- Contradictions (e.g., $0 = -3$) mean **no solution**.
- Identities (e.g., $0 = 0$) mean **infinite solutions**.

Thank You!

We appreciate your time and focus.

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